

CURRICULUM VITAE

Dr. Junke Wang

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EDUCATION

- PhD in Chemistry** – Eindhoven University of Technology, The Netherlands **2016 – 2020**
Dissertation: “Interfacial, Compositional and Morphological Engineering for Single- and Multi-junction Perovskite Solar Cells”
Supervisor: Prof. René A.J. Janssen
- MSc in Chemical Engineering** – Eindhoven University of Technology, The Netherlands **2014 – 2016**
Dissertation: “Sequential Deposition Method for Planar Perovskite Solar Cells”
With great appreciation
- BE in Materials Science and Engineering** – Central South University, China **2009 – 2013**
Cum Laude

ACADEMIC APPOINTMENTS

- Postdoc, Marie-Curie Fellow**, University of Oxford (Supervisor: Prof. Henry J. Snaith) **2023 – 2026**
- Lead research on hybrid wide-bandgap semiconductors for high-efficiency multi-junction solar cells
 - Secured £200,512 funding from the UKRI for Guarantee Funding for Marie-Curie Fellowship
- Postdoc**, jointly between University of Toronto (Prof. Edward H. Sargent) and Eindhoven University of Technology (Prof. René A.J. Janssen) **2020 – 2023**
- Developed stable high-bandgap perovskite materials for all-perovskite triple-junction solar cells
 - Conducted research on perovskite-based tandem photovoltaics for solar-driven water splitting
- PhD Candidate**, Eindhoven University of Technology (Supervisor: Prof. René A.J. Janssen) **2016 – 2020**
- Conducted research on hybrid semiconductors for solar energy storage, photodiodes, and light-driven CO₂ reduction
 - Developed prototype all-perovskite triple-junction solar cells
 - Supervised PhD/Master students and contributed to teaching undergraduate chemistry courses

RESEARCH INTERESTS

- Perovskite multijunction solar cells

- Light-trapping nanostructures for photovoltaics
- Operando characterization techniques for optoelectronics (XRD, XPS, TEM, CL, optical Spectroscopy)
- Sustainable materials and renewable energy technologies

AWARDS & GRANTS

Marie Curie Postdoctoral Fellowship (UKRI Guarantee, EP/Y029216/1, £200,512)	2023
The 2020 Chinese Government Award for Outstanding Self-financed Students Abroad (US\$6k)	2021
Amandus H. Lundqvist Scholarship Program, Eindhoven University of Technology, The Netherlands (~€36k)	2014

TEACHING EXPERIENCE

- Teaching assistant for third-year undergraduates in chemistry (Design-based Energy Course, 3 semesters, TU/e)
- Co-supervised 2 masters' thesis and 2 bachelors' thesis (TU/e)
 - Alessandro Caiazzo, dissertation "*Multidimensional perovskite solar cells*"
 - Hiten Tarpara, dissertation "*Pb-Sn based ideal-bandgap perovskite solar cell*"
 - Ronald de Bruijne, dissertation "*Fabrication of stable pure-phase FAPbI₃ perovskites for photovoltaic applications*"
 - Jasper Slingerland, dissertation "*Development of All-Bromide Pb-Sn Hybrid Wide Bandgap Perovskite Solar Cells*"

INTERNATIONAL CONFERENCES

Oral presentation at the MRS Fall Meeting 2025, United States	2025
Oral presentation at the TandemPV Workshop 2025, Belgium	2025
Invited talk at the Solar Energy Seminar at Helmholtz-Zentrum Berlin, Germany	2025
Oral presentation at the MRS Fall Meeting 2024, United States	2024
Oral presentation at the SPIE Optics + Photonics 2024, United States	2024
Oral presentation at the TandemPV Workshop 2024, The Netherlands	2024
Oral presentation at the 6 th International Conference on Perovskite Solar Cells and Optoelectronics, United Kingdom	2023
Oral presentation at the MRS Fall Meeting 2022, Virtual	2022
Oral presentation at the MRS Spring Meeting 2022, Virtual	2022
Oral presentation at the nanoGe Spring Meeting, Virtual	2021
Oral presentation at the 5 th International Conference on Perovskite Solar Cells and Optoelectronics, Switzerland	2019
Poster award at the International Conference on Hybrid and Organic Photovoltaics, Spain	2018

MEDIA OUTREACH

CNN.com	Razor-thin solar panels could be 'ink-jetted' onto your backpack or phone for cheap clean energy
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MSN.com	Solar energy breakthrough could mean solar panels will be a thing of the past
PV Magazine	All-perovskite triple-junction solar cell achieves 25.1% efficiency via new passivation technique
PV Magazine	Triple junction perovskite cell hits 16.8% efficiency

PUBLICATION LIST

- **Wang J.**,[†] Hu S.,[†] Chen Z.,[†] Snaith H.J. et al. *Exposing binding-favourable facets of perovskites for tandem solar cells.* Energy & Environmental Science, 2025, 18, 7680. <https://pubs.rsc.org/en/content/articlelanding/2025/ee/d5ee02462e>
- Hu S.,[†] **Wang J.**,[†] Wakamiya A., Snaith H.J. et al. *Steering tin–lead perovskite precursor solutions for multijunction photovoltaics.* Nature, 2025, 639, 93. <https://doi.org/10.1038/s41586-024-08546-y>
- **Wang J.**,[†] Branco B.,[†] Janssen R.A.J. et al. *All-perovskite tandem photovoltaics in light-driven electrochemical water splitting.* Nature Communications, 2025, 16, 174. <https://doi.org/10.1038/s41467-024-55654-4>
- **Wang J.**,[†] Zeng L.,[†] Zhang D.,[†] Maxwell A.,[†] Chen H.,[†] Janssen R.A.J., Sargent E.H. et al. *Halide homogenization for low energy loss in 2-eV-bandgap perovskites and increased efficiency in all-perovskite triple-junction solar cells.* Nature Energy, 2024, 9, 70. <https://www.nature.com/articles/s41560-023-01406-5>
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- Gómez P.,[†] **Wang J.**,[†] Más-Montoya M., Bautista D., Weijtens C.H.L., Curiel D., Janssen R.A.J. *Pyrene-based small molecular hole transport layers for efficient and stable narrow-bandgap perovskite solar cells.* Solar RRL, 2021, 5, 2100454. <https://onlinelibrary.wiley.com/doi/full/10.1002/solr.202100454>
- **Wang J.**, Zardetto V., Datta K., Zhang D., Wienk M.M., Janssen R.A.J. *16.8% Monolithic All-perovskite Triple-junction Solar Cells via A Universal Two-step Solution Process.* Nature Communications, 2020, 11, 5254. <https://www.nature.com/articles/s41467-020-19062-8>
- **Wang J.**, Datta K., Li J., Verheijen M.A., Zhang D., Wienk M.M., Janssen R.A.J. *Understanding the Film Formation Kinetics of Sequential Deposited Narrow-Bandgap Pb–Sn Hybrid Perovskite Films.* Advanced Energy Materials, 2020, 10, 2000566. <https://onlinelibrary.wiley.com/doi/full/10.1002/aenm.202000566>
- **Wang J.**, Datta K., Weijtens C.H.L., Wienk M.M., Janssen R.A.J. *Insights into fullerene passivation of SnO₂ electron transport layers in perovskite solar cells.* Advanced Functional Materials, 2019, 29, 1905883. <https://onlinelibrary.wiley.com/doi/full/10.1002/adfm.201905883>
- **Wang J.** et al. *Highly Efficient Perovskite Solar Cells Using Non-Toxic Industry Compatible Solvent System.* Solar RRL, 2017, 1, 1700091. <https://onlinelibrary.wiley.com/doi/full/10.1002/solr.201700091>
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References

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